9. What is microbial biotransformation? Clarify.

10. Briefly elucidate about PCR.

11. Discuss in short about storage conditions and stability of officials vaccines.

12. Give details of design of a fermenter and its various controls.

13. Write in short about biosensors.

Printed Pages— 10

Roll No. .....

341655(41)

B. Pharmacy (Sixth Semester) Examination,
April-May 2020

(PCI Scheme)

PHARMACEUTICAL BIOTECHNOLOGY

THEORY—(BP605T)

Time Allowed : Three hours

Maximum Marks : 75

Note *: Attempt all parts as directed.*

Part-A **1×20=20**

Note *: Attempt all questions.*

1. Multiple Choice Questions :

(i) What is the clinical application of monoclonal antibodies?

(a) Biosensors

(b) Transplant rejection

(c) Infectious disease

(d) Purification of drugs

(ii) Restriction endonucleases are most widely used in recombinant DNA technology. They are obtained from :

(a) Bacteriophages

(b) Bacterial cells

(c) Plasmids

(d) All prokaryotic cell

(iii) Sterilization of tissue culture medium is done by :

(a) Mixing the medium with antifungal agents

(b) Filtering the medium through fine sieve

(c) Autoclaving of medium at 120° for 15 min

(d) Keeping the medium at –20°C

(iv) Micropropagation is a technique :

(a) For production of many plants that are clones of each other

(b) For production of haploid plant

(c) For production of somatic hybrids

(d) For production of somaclonal plants

(v) Plasmids are suitable vectors for gene cloning because :

(a) These can shuttle between prokaryotic and eukaryotic cells https://www.csvtuonline.com

(b) These are small circular DNA molecules with their own replication origin site

(c) These are small circular DNA molecules, which can integrate with host chromosomal DNA

(d) These often carry antibiotic resistance genes

(vi) Double stranded DNA denaturation with specified limit of temperature is :

(a) Reversible reaction

(b) Irreversible reaction

(c) Either (a) or (b)

(d) None of these

(vii) In a PCR reaction after four cycles, each molecule of a duplex DNA give rise to :

(a) 16 double stranded DNA

(b) 16 single strands of DNA

(c) 18 single stranded of DNA

(d) 18 double stranded DNA

(viii) The PCR, polymerase chain reaction is becoming the method of choice for :

(a) Alteration of gene

(b) Screening gene

(c) Sterilization of gene

(d) All of these

(ix) The first immunoglobulin heavy chain class to be expressed on the surface of a newly produced B-cell is :

(a) IgA

(b) IgD

(c) IgG

(d) IgM

(x) The cytoplasmic region of surface IgM consists of :

(a) A single H chain constant region domain

(b) A light chain

(c) 110 amino acids

(d) 3 amino acids

(xi) Which of the following is considered an autoimmune disease :

(a) Rheumatoid arthritis

(b) AIDS

(c) SCID

(d) Agammaglobulinemia

(xii) A transplant between individuals of different animal species is termed as :

(a) Allograft

(b) Isograft

(c) Enterograft

(d) Xenograft

(xiii) HIV has a high mutation rate due to the imprecise operation of its :

(a) Viral membrane

(b) Reverse transcriptase

(c) Protease

(d) Dismutase

(xiv) Electroporation is also used for taking up the DNA by the cell. It constitutes of ..

(a) Inserting the DNA into the cells via an electric shock

(b) Increased efficiency than both natural and chemical methods

(c) Causing the least amount of damage in comparison to other methods

(d) Decreased efficiency than both natural and chemical methods

(xv) Which of the statements hold true for conjugation?

(a) Conjugation is the natural process of transferring DNA from one species to another

(b) It is the artificial process in case the cells are not able to take them up naturally

(c) The plasmids are transferred from one cell to another by physical contact

(d) The plasmids are transferred from one cell to another by chemical means

(xvi) The addition or deletion of a nucleotide base pair involves :

(a) Point mutation

(b) Silent mutation

(c) Nonsense mutation

(d) Frame shift mutation

(xvii) The final electron acceptor in lactic acid fermentation is :

(a) Oxygen

(b) Lactic acid

(c) Pyruvate

(d) NAD

(xviii) Pasteur effect discovered in 1857, is :

(a) Inhibiting effect of oxygen on the fermentation process

(b) Aerating yeasted broth causes yeast cell growth to decrease, while conversely, fermentation rate increase

(c) (a) and (b)

(d) All of these

(xix) EFB class 4 consists of ..... :

(a) Low-risk microorganisms

(b) High-risk microorganisms

(c) Medium-risk microorganisms

(d) Environmental-risk microorganisms

(xx) Which of the following fermenters are characterized by height to diameter ratio :

(a) Tower fermenter

(b) Airlift fermenter

(c) Hollow fibre

(d) Perfusion bioreactor

Part-B

Long Answer Questions 2×10=20

Note : *Attempt any **two** questions.*

2. Describe methods of enzyme immobilization and applications.

3. Explain in detail application of r-DNA technology and genetic engineering in the production of hepatitis vaccine.

4. Discuss and illustrate production of penicillin by fermentation technology.

Part-C

Short Answer Questions 7×5=35

Note : *Attempt any **seven** questions.*

5. Write a note on types of mutants.

6. Explain southern blotting.

7. What is hybridoma technology? Discuss in brief.

8. Define immunity and discuss types of immunity in short.